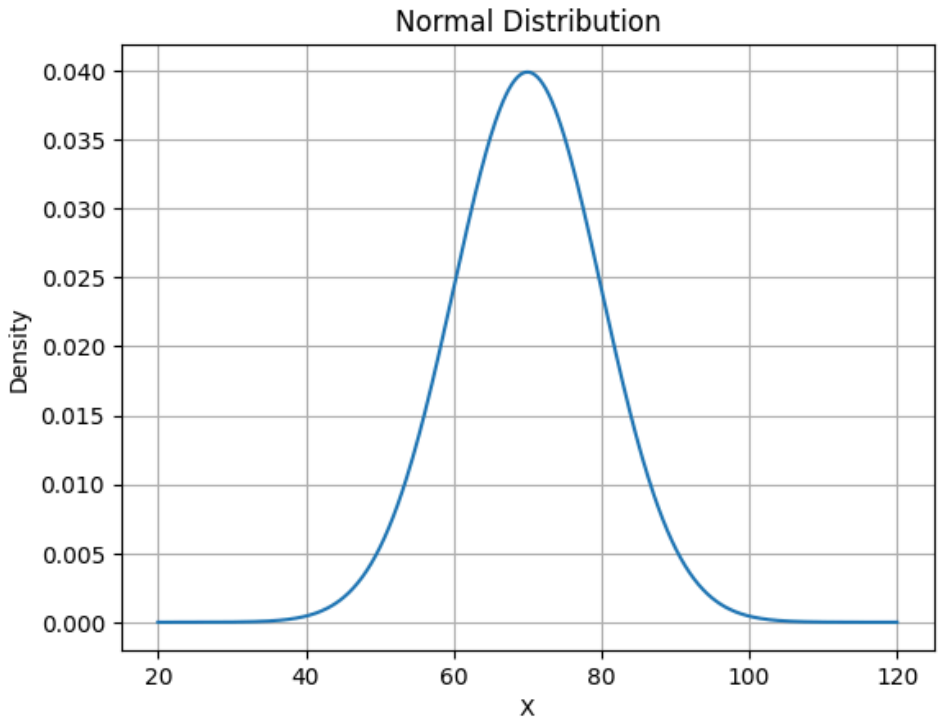
```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
```

Q1

```
mu = 70
sigma = 10
x = np.linspace(20, 120, 500)
pdf = stats.norm.pdf(x, loc=mu, scale = sigma)
plt.plot(x,pdf)
plt.xlabel("X")
plt.ylabel("Density")
plt.title("Normal Distribution")
plt.grid()
plt.show()
```



Q1)

The distribution is symmetric.

The curve is centered around 70.

If standard deviation is increased, the spread of the distribution increases.

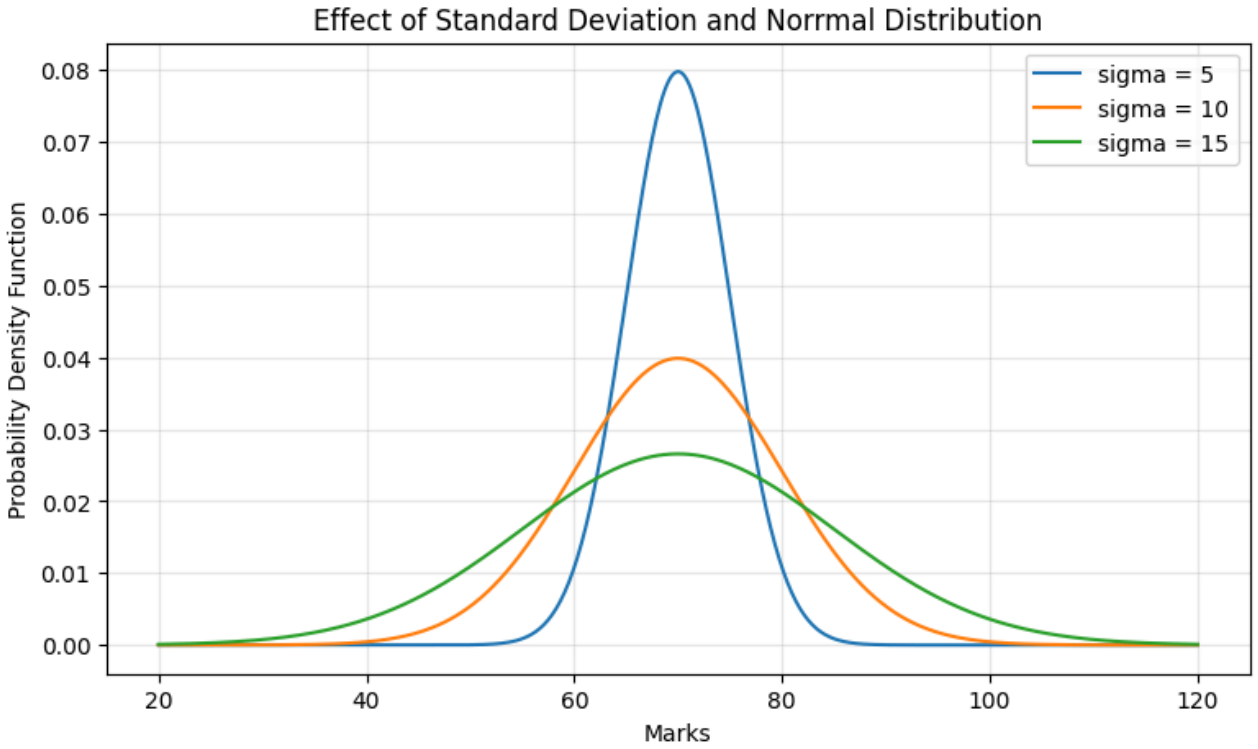
Q2)

```
mu = 70
x = np.linspace(20, 120, 500)
pdf1 = stats.norm.pdf(x, loc=mu, scale = 5)
pdf2 = stats.norm.pdf(x, loc=mu, scale = 10)
pdf3 = stats.norm.pdf(x, loc=mu, scale = 15)

plt.figure(figsize=(9,5))

plt.plot(x, pdf1, label="sigma = 5")
plt.plot(x, pdf2, label="sigma = 10")
plt.plot(x, pdf3, label="sigma = 15")

plt.xlabel("Marks")
plt.ylabel("Probability Density Function")
plt.title("Effect of Standard Deviation and Norrmal Distribution")
plt.legend()
plt.grid(alpha = 0.3)
plt.show()
```



Q2)

i) The distribution with sigma = 5 is the most concentrated. ii) The distribution with sigma = 15 has the greatest spread. iii) As the standard deviation increases, the peak height of the curve decreases.

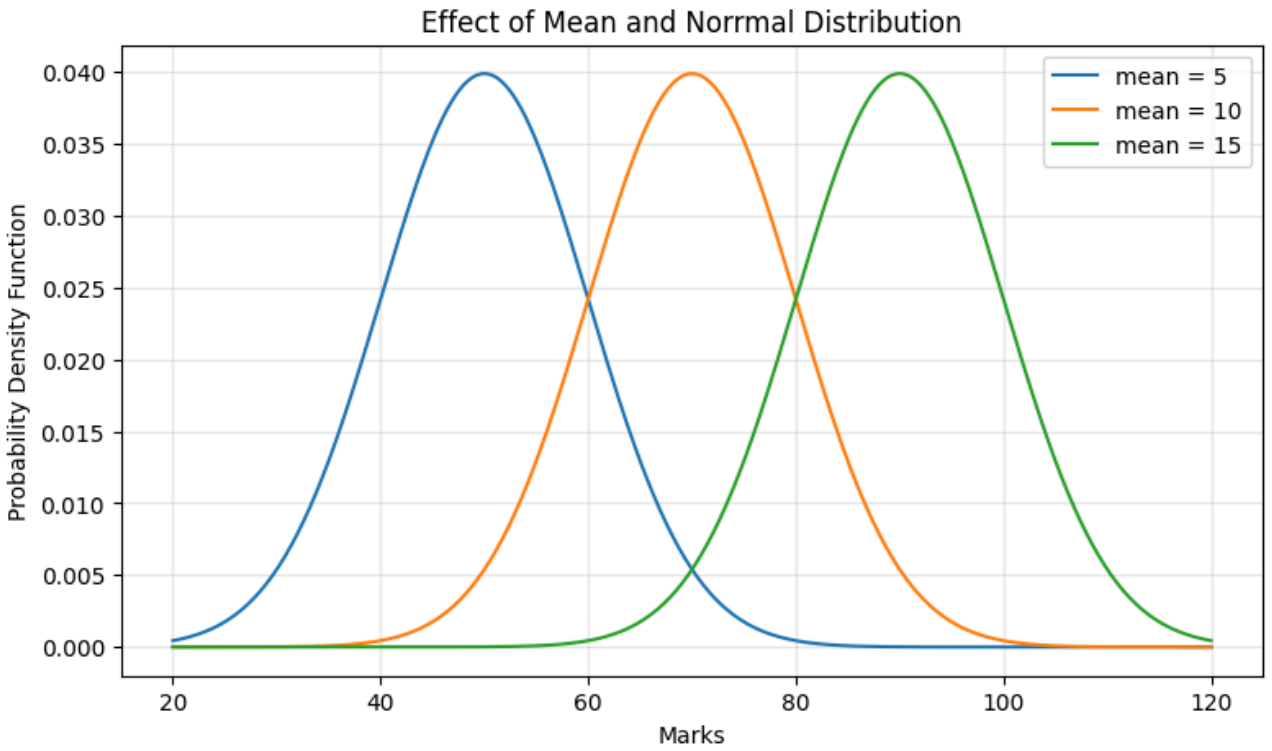
Q3)

```
sigma= 10
x = np.linspace(20, 120, 500)
pdf1 = stats.norm.pdf(x, loc=50, scale = sigma)
pdf2 = stats.norm.pdf(x, loc=70, scale = sigma)
pdf3 = stats.norm.pdf(x, loc=90, scale = sigma)

plt.figure(figsize=(9,5))

plt.plot(x, pdf1, label="mean = 5")
plt.plot(x, pdf2, label="mean = 10")
plt.plot(x, pdf3, label="mean = 15")
```

```
plt.xlabel("Marks")
plt.ylabel("Probability Density Function")
plt.title("Effect of Mean and Norrmal Distribution")
plt.legend()
plt.grid(alpha = 0.3)
plt.show()
```



Based on the plot generated for Q3, when the mean changes while the standard deviation remains constant, the normal distribution shifts horizontally along the x-axis. Its shape, including its spread and height, remains unchanged.

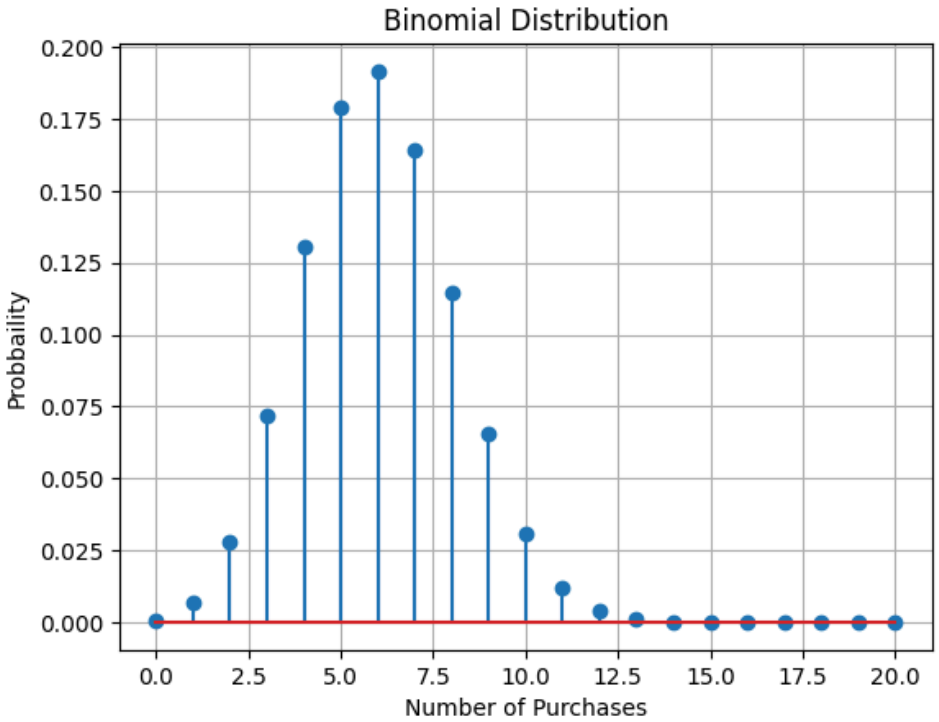
Q4)

```
import numpy as np

n = 20
p = 0.3

x = np.arange(0, n+1)
pmf = stats.binom.pmf(x, n ,p)
```

```
plt.stem(x, pmf)
plt.xlabel("Number of Purchases")
plt.ylabel("Probbaility")
plt.title("Binomial Distribution")
plt.grid()
plt.show()
```

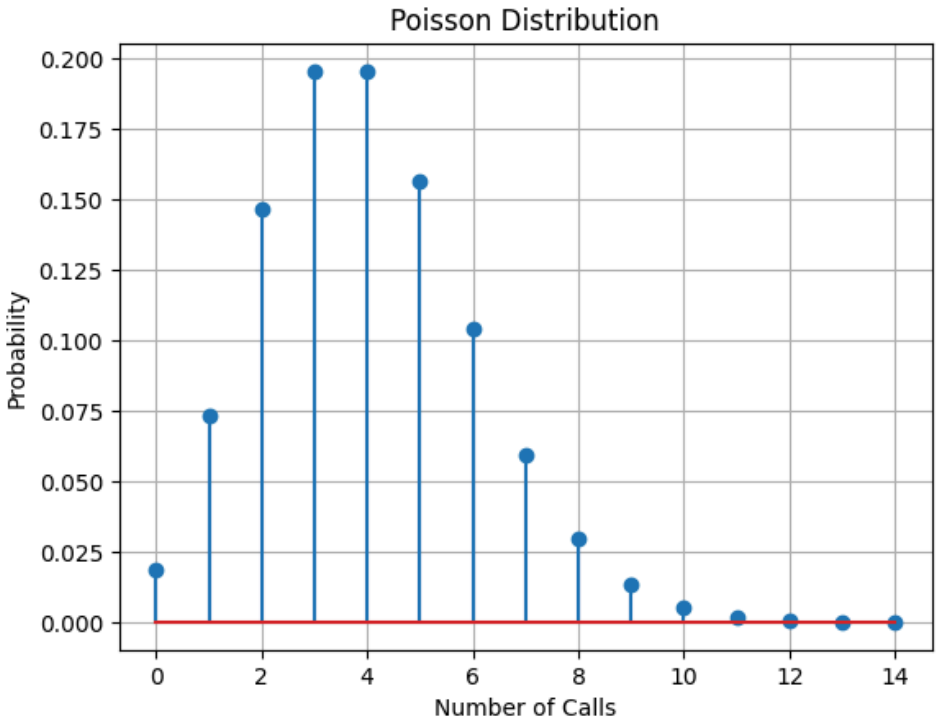


For Q4, the plot above shows the Probability Mass Function (PMF) of the binomial distribution for 20 customers with a probability of purchase of 0.3. The x-axis represents the number of purchases, and the y-axis represents the probability of that number of purchases occurring. The plot visually represents the likelihood of different numbers of purchases among the 20 customers.

Q5)

```
lam = 4
x = np.arange(0, 15)
pmf = stats.poisson.pmf(x, mu=lam)
plt.stem(x, pmf)
plt.xlabel("Number of Calls")
plt.ylabel("Probability")
plt.title("Poisson Distribution")
```

```
plt.grid()
plt.show()
```



For Q5:

The value of lambda (average rate of events) is 4 (as defined in the code: lam = 4). The Poisson distribution is discrete. The expected number of calls per minute is equal to the value of lambda, which is 4 calls per minute.